

Practice Test — Chapter 9: Motion

Year 10 Science (Victorian Curriculum, VCSSU133)

Name: _____ Date: _____

Time allowed: 50 minutes Total marks: /50

Instructions: Answer all questions. Show full working for calculations and include units. Use $g = 9.8 \text{ m/s}^2$. The answer key is at the end — try the whole test before checking.

Useful formulas: $v = d \div t$ | $a = (v_2 - v_1) \div t$ | $F = ma$ | $W \text{ (weight)} = mg$ | $1 \text{ m/s} = 3.6 \text{ km/h}$

Section A — Multiple choice (10 marks, 1 mark each)

Circle the best answer.

1. Which quantity includes **direction** as well as size? - A. distance - B. speed - C. displacement - D. time
 2. The correct unit for acceleration is: - A. m/s - B. m/s^2 - C. km/h - D. N
 3. To convert a speed from m/s to km/h you: - A. multiply by 3.6 - B. divide by 3.6 - C. multiply by 1000 - D. divide by 3600
 4. On a **distance-time** graph, the gradient (slope) of the line represents the: - A. acceleration - B. displacement - C. speed - D. force
 5. On a **speed-time** graph, the area under the line represents the: - A. acceleration - B. distance travelled - C. average speed - D. force
 6. A horizontal line on a **speed-time** graph means the object is: - A. stationary - B. accelerating - C. travelling at constant speed - D. decelerating
 7. Newton's first law of motion is also known as the law of: - A. gravity - B. inertia - C. acceleration - D. action-reaction
 8. If the net force on a moving object is **zero**, the object will: - A. speed up - B. slow down and stop - C. continue at a constant speed in a straight line - D. change direction
 9. Which equation represents Newton's second law of motion? - A. $W = Fs$ - B. $F = ma$ - C. $v = d \div t$ - D. $Ug = mgh$
 10. The weight of an object is best described as: - A. the amount of matter in the object - B. the force of gravity acting on the object - C. the object's mass in kilograms - D. the object's speed
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Section B — Short answer (14 marks)

11. Distinguish between **distance** and **displacement**. (2 marks)
12. Distinguish between **speed** and **velocity**. (2 marks)

- 13.** Distinguish between **average speed** and **instantaneous speed**, giving an example of how each is measured. (2 marks)
- 14.** Define **inertia** and give an everyday example. (2 marks)
- 15.** State Newton's **three** laws of motion. (3 marks)
- 16.** Explain the difference between **mass** and **weight**. (2 marks)
- 17.** Explain why an astronaut's **weight** is different on the Moon but their **mass** stays the same. (1 mark)
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Section C — Calculations and graphs (18 marks)

Show all working and include units.

- 18.** A runner completes a 100 m race in 12.5 s. Calculate their **average speed**. (2 marks)
- 19.** A car is travelling at 90 km/h. Convert this speed to **m/s**. (2 marks)
- 20.** A dirt bike accelerates from rest (0 m/s) to 15 m/s in 3 seconds. Calculate its **acceleration**. (3 marks)
- 21.** A force of 138 N is applied to an object of mass 6 kg. Calculate its **acceleration**. (3 marks)
- 22.** Calculate the **weight** (force due to gravity) of a 70 kg person on Earth ($g = 9.8 \text{ m/s}^2$). (2 marks)
- 23.** A person walks **4 m east** and then **3 m north**. (3 marks)
- a) What is the total **distance** travelled? (1 mark)
 - b) Use Pythagoras' theorem to find the **size of the displacement**. (2 marks)
- 24.** Look at the speed-time graph description below: an object speeds up steadily from 0 to 10 m/s over the first 5 seconds, then travels at a constant 10 m/s for the next 3 seconds. (3 marks)
- a) Calculate the acceleration during the first 5 seconds. (1 mark)
 - b) Calculate the distance travelled during the constant-speed section (5 s to 8 s). (2 marks)
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Section D — Extended response (8 marks)

- 25.** Using Newton's laws, explain what happens to a passenger's body in a car that suddenly brakes, and explain how a **seatbelt** keeps the passenger safe. Refer to **inertia** and **force** in your answer. (4 marks)
- 26.** A skydiver jumps from a plane and eventually falls at a constant speed called *terminal velocity*. (4 marks)
- a) Name the two main forces acting on the skydiver and their directions. (2 marks)

- b) Using Newton's first law, explain why the skydiver falls at a **constant** speed at terminal velocity (even though they are still falling). (2 marks)

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ANSWER KEY — Motion

Section A 1. C — displacement 2. B — m/s^2 3. A — multiply by 3.6 4. C — speed 5. B — distance travelled 6. C — travelling at constant speed 7. B — inertia 8. C — continue at a constant speed in a straight line 9. B — $F = ma$ 10. B — the force of gravity acting on the object

Section B

11. Distance = the total length of path travelled (size only). Displacement = how far and in which direction an object is from its starting point (size and direction). (2)

12. Speed = how fast something travels (distance $\div$ time, size only). Velocity = how fast in a particular direction (displacement $\div$ time, has direction). (2)

13. Average speed = total distance $\div$ total time over a whole journey (e.g. timing a runner over 100 m). Instantaneous speed = speed at a single instant (e.g. a car speedometer or speed camera). (2)

14. Inertia = an object's tendency to resist a change in its motion (stay at rest or keep moving the same way). Example: your body lurching forward when a car suddenly brakes. (2)

15. First: an object stays at rest or moves at constant speed in a straight line unless acted on by an unbalanced force. Second: net force = mass $\times$ acceleration ($F = ma$). Third: for every action there is an equal and opposite reaction force. (3)

16. Mass = the amount of matter in an object (kg), unchanged by location. Weight = the force of gravity acting on the object (N), which depends on the gravitational field strength. (2)

17. The Moon's gravity is weaker ($g = 1.6 \text{ m/s}^2$ vs 9.8 m/s^2), so the weight force is smaller, but the amount of matter (mass) is unchanged. (1)

Section C

18. $v = d \div t = 100 \div 12.5 = \mathbf{8 \text{ m/s}}$ (2)

19. $90 \div 3.6 = \mathbf{25 \text{ m/s}}$ (2)

20. $a = (v_2 - v_1) \div t = (15 - 0) \div 3 = \mathbf{5 \text{ m/s}^2}$ (3)

21. $a = F \div m = 138 \div 6 = \mathbf{23 \text{ m/s}^2}$ (3)

22. $W = mg = 70 \times 9.8 = \mathbf{686 \text{ N}}$ (2)

23. a) distance = $4 + 3 = \mathbf{7 \text{ m}}$ (1) b) displacement = $\sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = \mathbf{5 \text{ m}}$ (2) (*direction $\approx 37^\circ$ north of east for full vector*)

24. a) $a = (10 - 0) \div 5 = \mathbf{2 \text{ m/s}^2}$ (1) b) distance = speed $\times$ time = $10 \times 3 = \mathbf{30 \text{ m}}$ (area of the rectangle under the graph) (2)

Section D

25. Marking points (4): When the car brakes, a force stops the car, but the passenger's body has **inertia** (Newton's first law) and continues moving forward at the original speed. Without a restraint the body would keep moving until a force stopped it (e.g. the windscreen). The **seatbelt applies the stopping force** to the body, and by stretching slightly it increases the stopping time/distance, which **reduces the force** on the passenger and lowers injury. (*Award for: inertia, body continues forward, seatbelt applies force, reduces force by increasing time/distance.*)

26. a) Weight (force of gravity) acting **downwards**; air resistance (drag/friction) acting **upwards**. (2) b) (2) At terminal velocity the upward air resistance has grown until it **balances** the downward weight, so the **net force is zero**. By Newton's first law, with balanced forces there is no acceleration, so the skydiver continues at a **constant speed** (but still moving).

END OF TEST